Lecture 11b
SGAM & comms

Geert Deconinck

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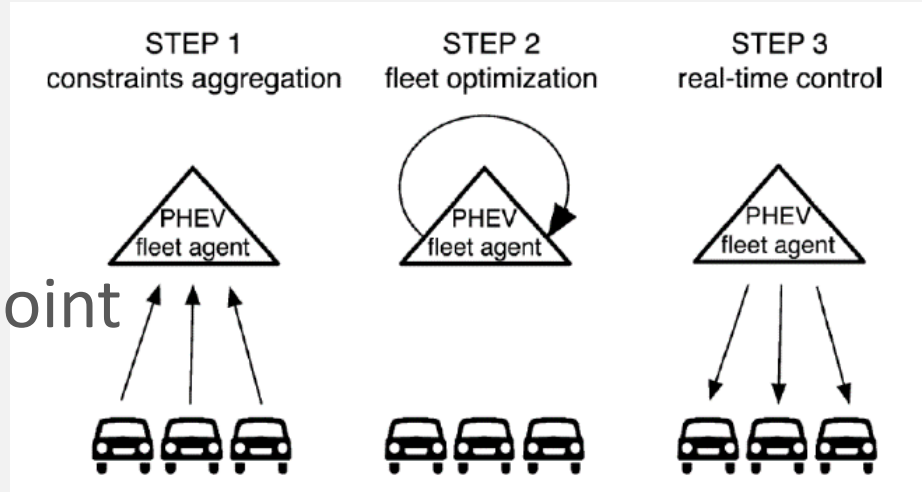
Outline

Three step approach for flexibility management

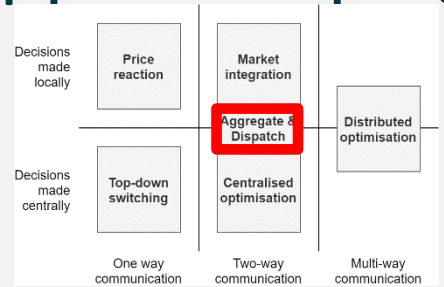
- SGAM - smart grid architecture model
  - structure
  - example
  - layers' contents

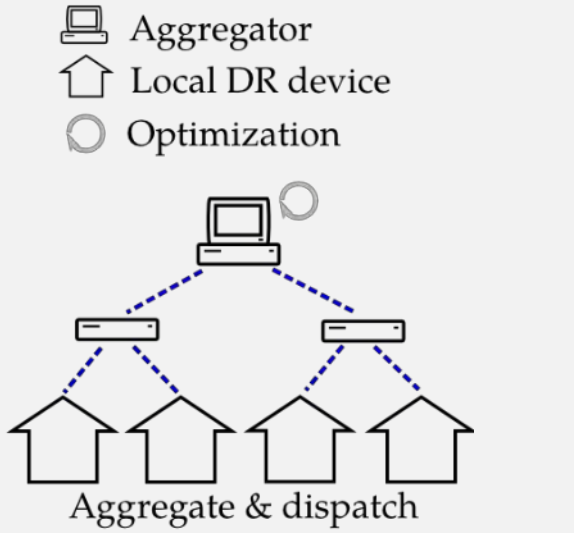
Three-step approach for clusters of flex providers

- three-step approach (TSA)
  - 1 aggregate flexibility and constraints
  - 2 centrally determine optimal aggregated set point
  - 3 dispatch set point to individual devices



- applied to plug-in hybrid electric vehicle charging

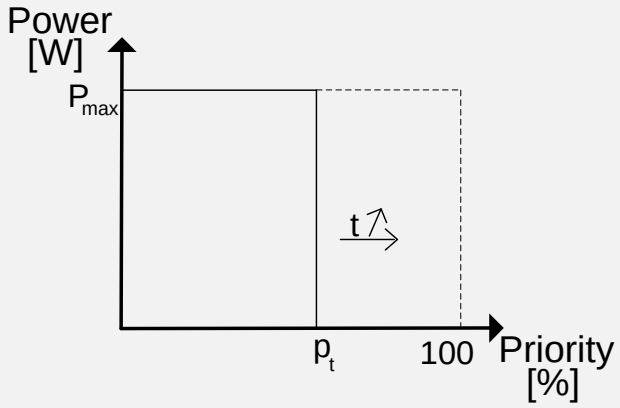




- S. Vandael, B. Claessens, M. Hommelberg, T. Holvoet, G. Deconinck, "A scalable three-step approach for demand side management of plug-in hybrid vehicles," *IEEE Trans. on Smart Grid*, No. 4(2): 720-728, 2013.

TSA – step 1: constraint aggregation

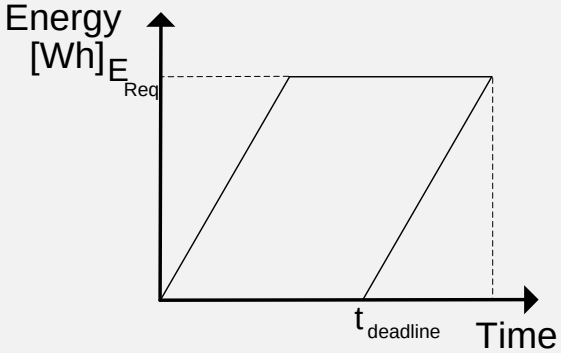
- power flexibility



$$b_{t,i}(p) = P_i^{\max}(1 - H(p - p_{t,i})) \quad \forall p \in \{0, ..., 1\}$$

$$b_i(p) = \begin{cases} P_i^{\max} & 0 \leq p^* \leq p_i - 1/\gamma \\ \gamma P_i^{\max}(p_i - p) & p_i - 1/\gamma < p^* < p_i \\ 0 & p_i \leq p^* \end{cases}$$

- energy flexibility

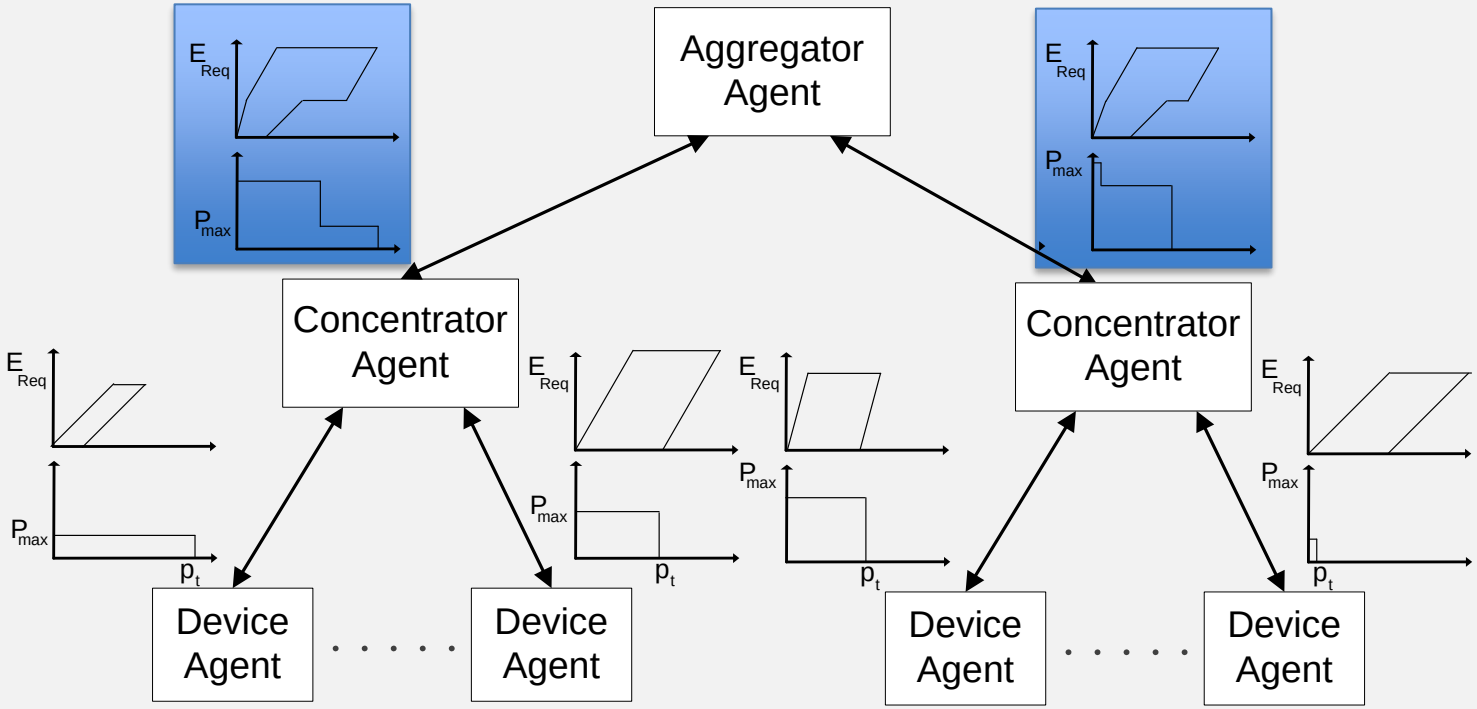


$$^i\mathbf{E}^{\max} = \left\{ {}^iE_t^{\max} | {}^iE_t^{\max} = \min({}^iP^{\max}t, {}^iE^{\text{req}}), \quad \forall t \in \{0, \ldots, {}^it_{\text{dep}}\} \right\}$$

$$^i\mathbf{E}^{\min} = \left\{ {}^iE_t^{\min} | {}^iE_t^{\min} = \max\left({}^iE^{\text{req}} - {}^iP^{\max}({}^it_{\text{dep}} - t), 0\right) \quad \forall t \in \{0, \ldots, {}^it_{\text{dep}}\} \right\}$$

TSA – step 1: constraint aggregation

- power $\mathbf{P}_i^{\text{dem}} = \{P_{i,p}^{\text{dem}} | \, P_{i,p}^{\text{dem}} = b_i(p), \forall p \in \{0, ..., 1\}\}$
- energy $\mathbf{E}^{\max} = \sum_{i=1}^{N} \mathbf{E}_i^{\max}, \; \mathbf{E}^{\min} = \sum_{i=1}^{N} \mathbf{E}_i^{\min}, \; \mathbf{P}^{\text{dem}} = \sum_{i=1}^{N} \mathbf{P}_i^{\text{dem}}$

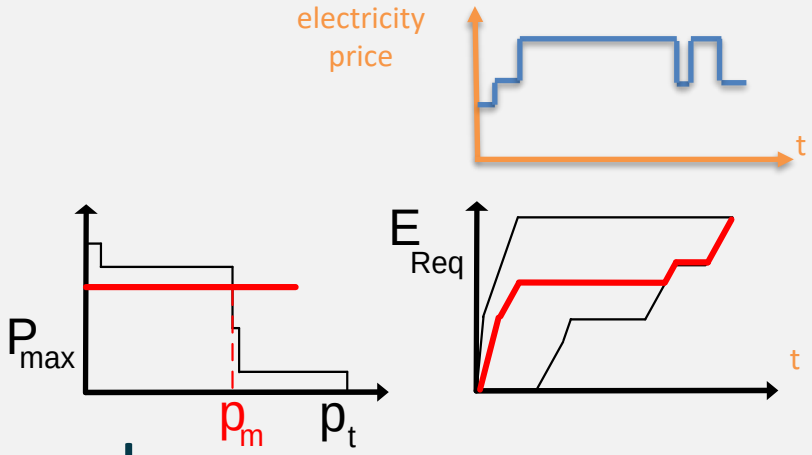


TSA – step 2: derive optimal charging schedule

- optimisation problem

$$\mathbf{P}^* = \min_{\mathbf{P}\in\mathbb{R}^T} f(\mathbf{P})$$

$$\text{s.t.}: \quad 0 \le P_t \le P^{\max} \quad \forall t \in \mathcal{T}$$
$$E_t^{\min} \le E_t \le E_t^{\max} \quad \forall t \in \mathcal{T}$$
$$E_{t+1} = E_t + P_t \Delta t \quad \forall t \in \mathcal{T}$$



- cost minimisation for determining optimal demand

$$f(\mathbf{P}) = \sum_{t=1}^{T} \lambda_t P_t \Delta t$$

  - P_t = cluster power consumption
  - λ_t = real time pricing tariff

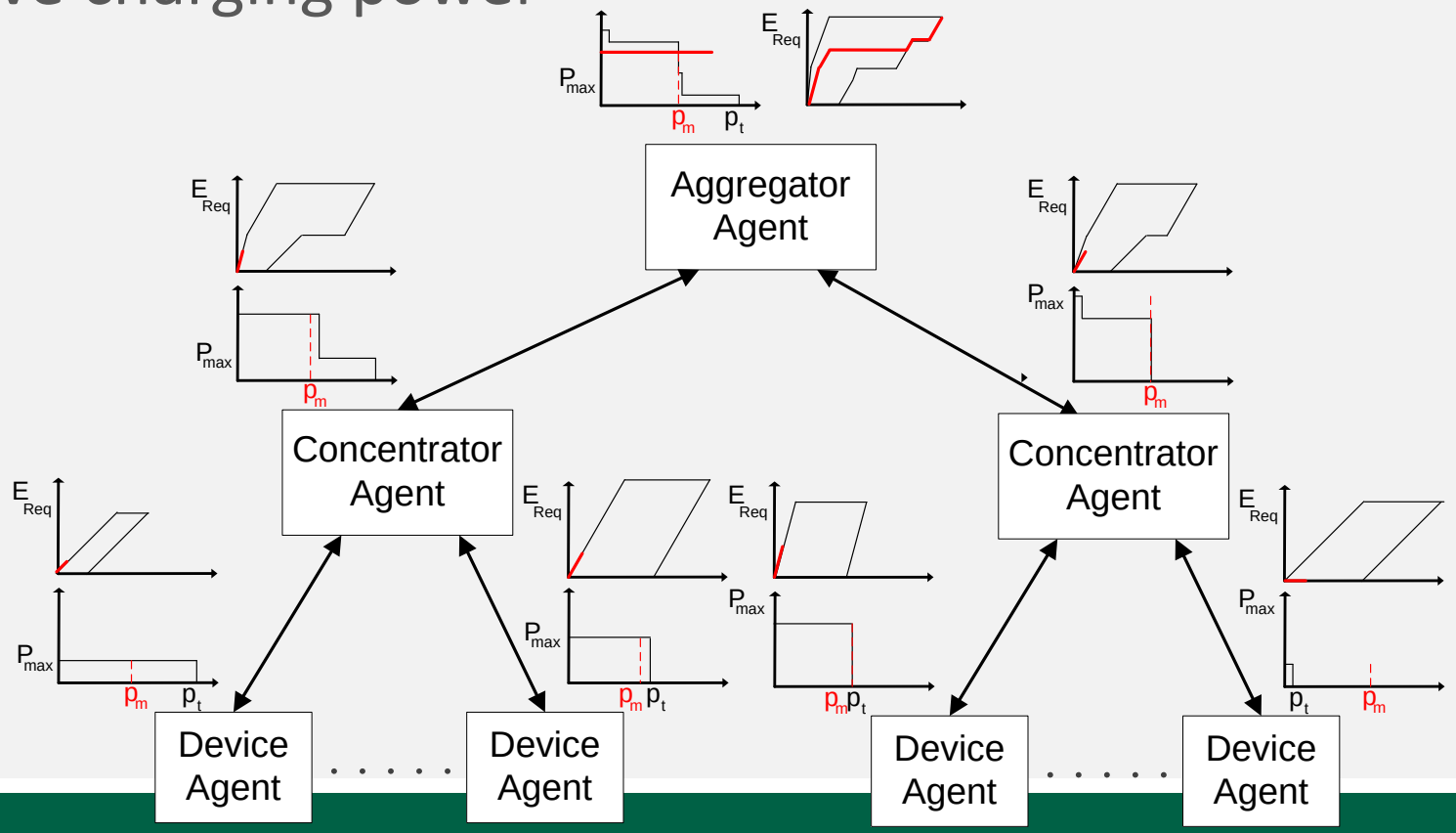
Aggregator Agent

- translate optimal demand P_1 to market priority p^*

$$p^* = \min_{p^*\in\{0,\dots,1\}} \left| \sum_{n=1}^{N} \mathbf{P}_i^{\text{dem}} - P_1^* \right|$$

TSA – step 3: real-time control

- dispatch priority
  - → all aggregators and device agents know priority
  - and can derive charging power



TSA - results

- savings

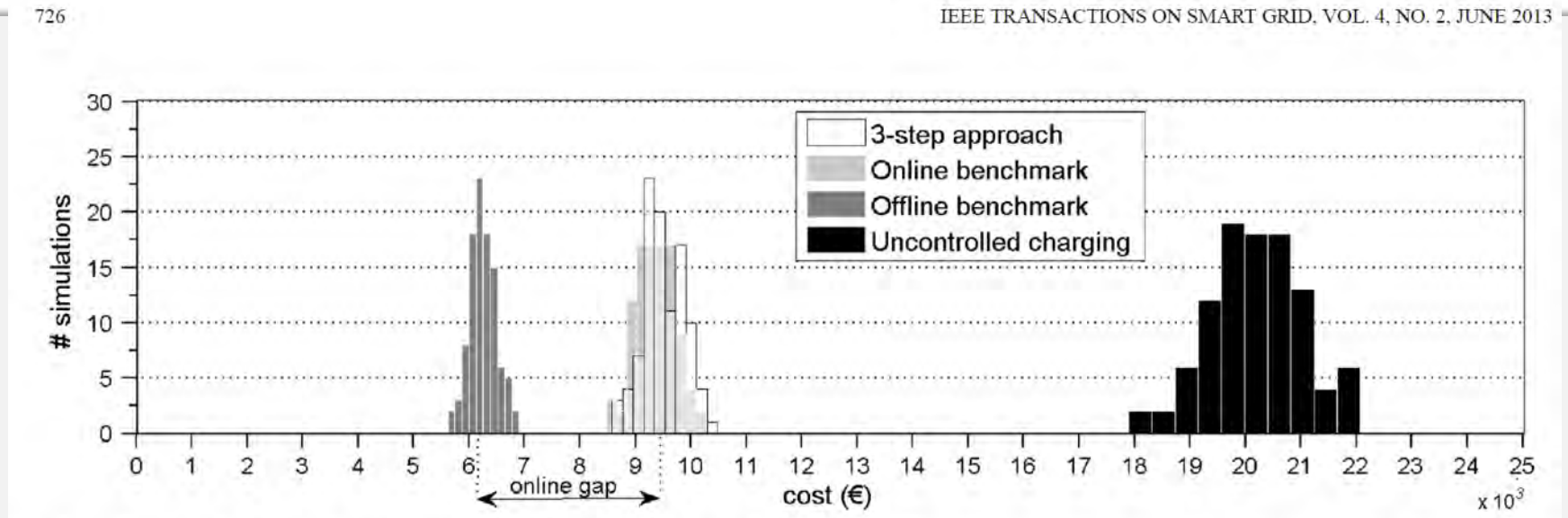


Fig. 7. Distribution of the total cost for each solution as result of a Monte-Carlo experiment.

- scalability

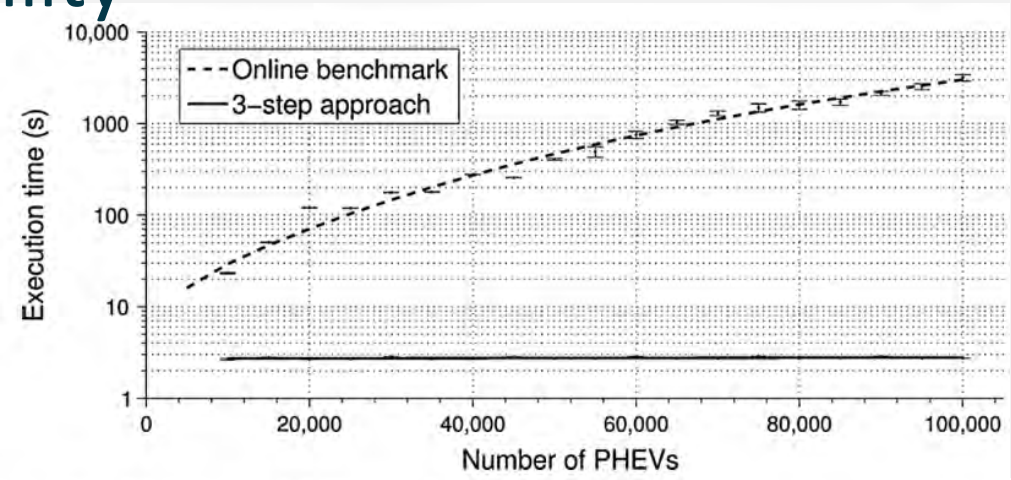


Fig. 11. Vertical scalability analysis.

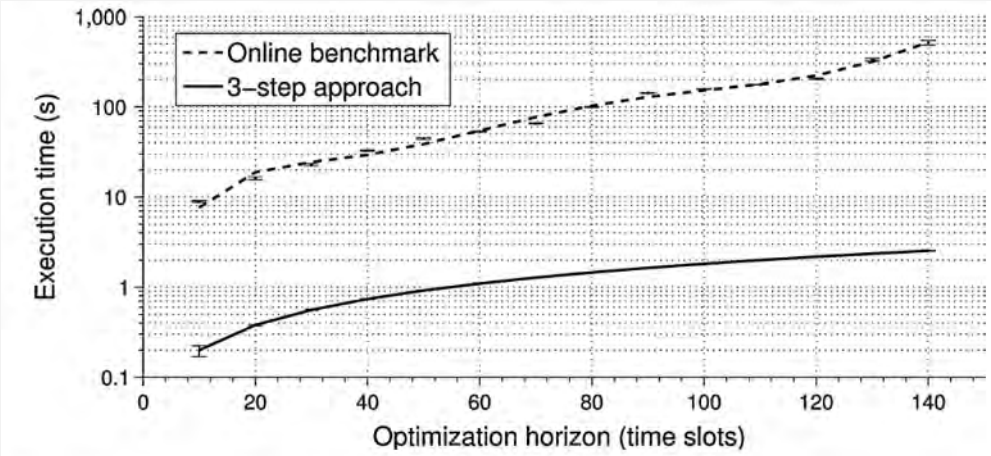


Fig. 12. Horizontal scalability analysis.

Outline

- Three step approach for flexibility management
- SGAM - smart grid architecture model
  - structure
  - example
  - layers' contents

Smart grid reference architecture model (SGAM)

- CEN/CENELEC/ETSI Smart Grid Coordination Group [SG-CG], "Smart grid reference architecture", Nov 2012, 107 pages
  http://ec.europa.eu/energy/gas_electricity/smartgrids/doc/xpert_group1_reference_architecture.pdf

- interoperability [IEC61850-2010]
  - ability of two or more devices from the same vendor, or different vendors, to exchange information and to use that information for correct cooperation.

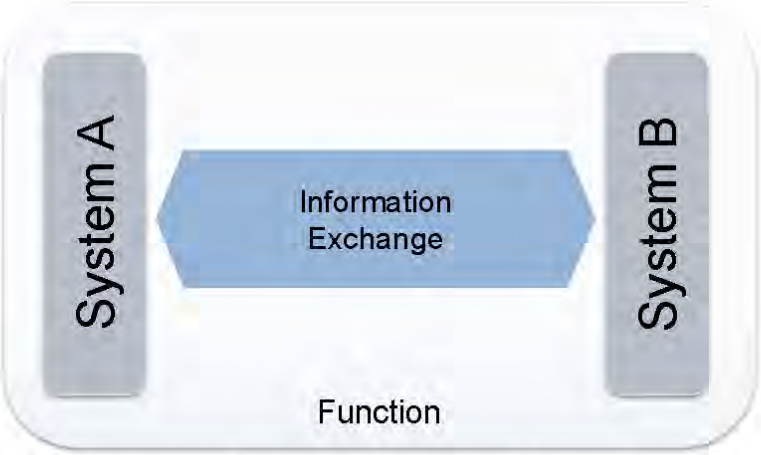


Figure 3: Definition of interoperability – interoperable systems performing a function

SGAM - smart grid architecture model

- goals: 3D representation of smart grid applications
  - z-axis: interoperability layers
  - x-axis: domains
  - y-axis: zones
- reference architectures

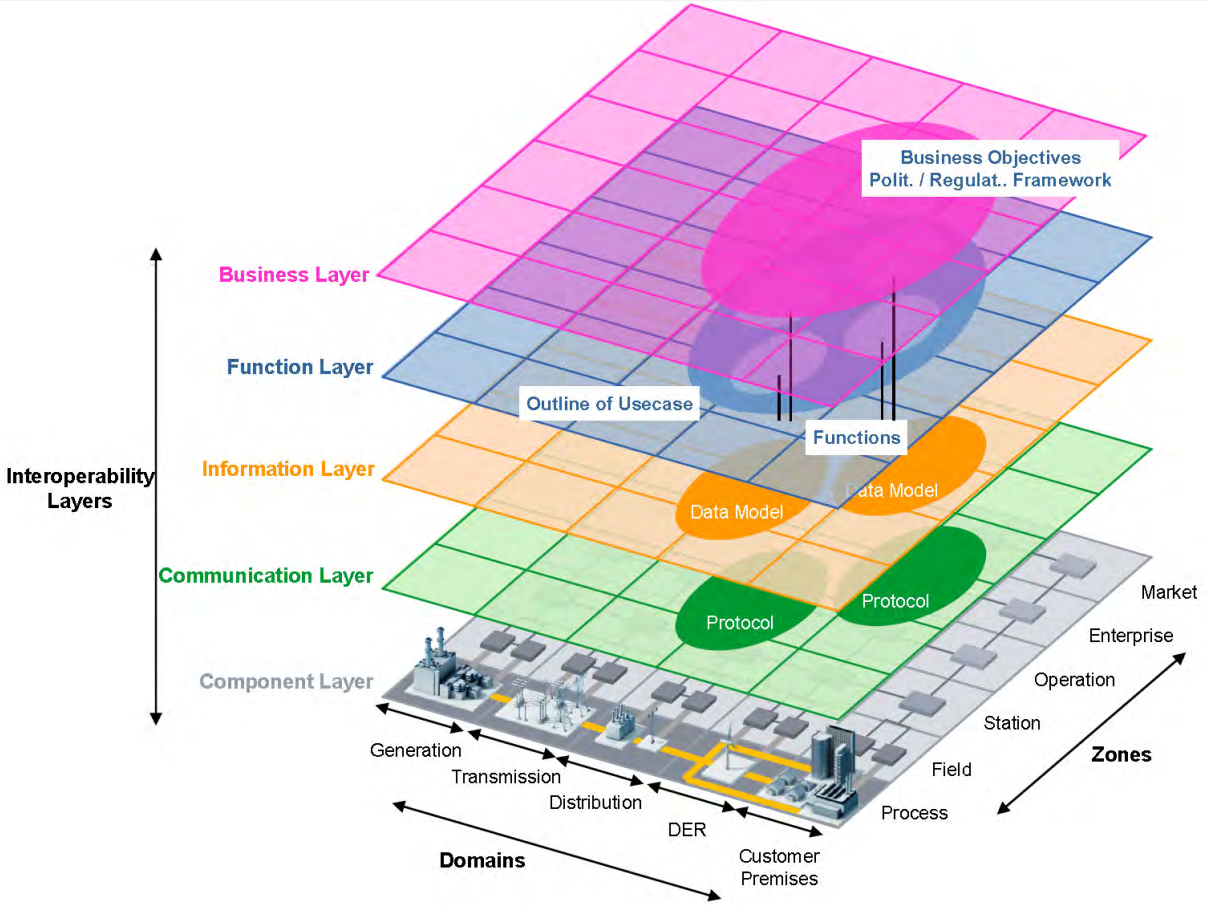


Figure 8: SGAM framework

z-axis: SGAM interoperability layers

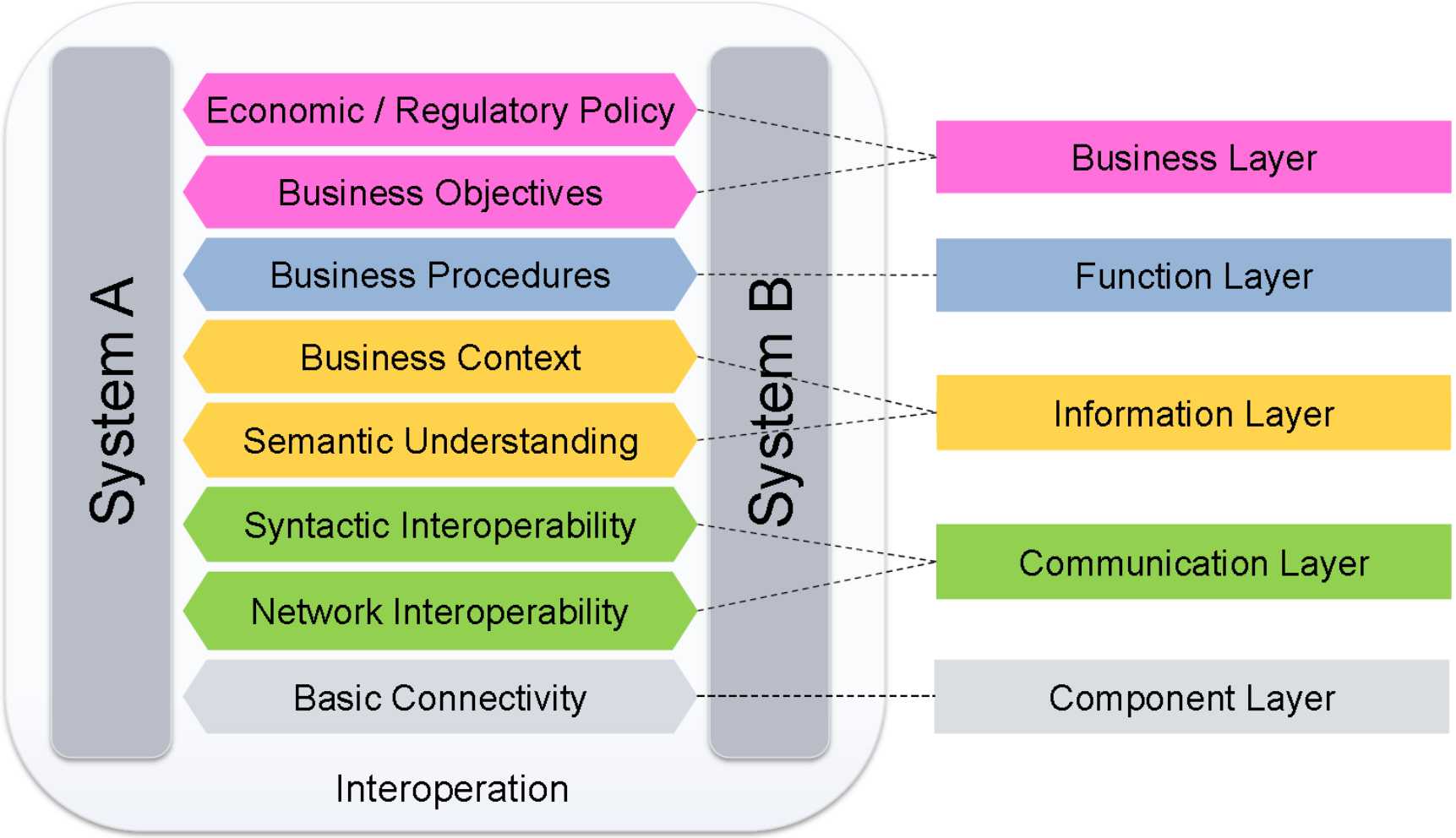
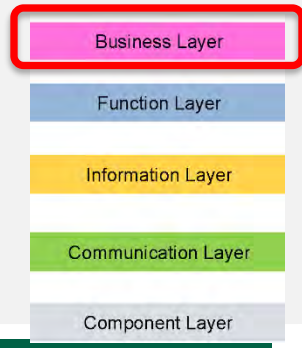


Figure 6: Grouping into interoperability layers

Business Layer

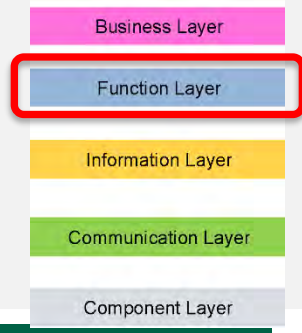
- represents business view on info exchange related to smart grids
- can be used to map regulatory and economic (**market) structures and policies**, **business models**, business **portfolios** (products & services) of market parties involved; also, business capabilities and **business processes**
- supports *business executives* in decision making related to (new) **business models** and specific business projects (**business case**) as well as *regulators* in defining new **market models**
- *e.g. trading aggregated flexibility on day-ahead market (TSA – three step approach)*



Function Layer

- describes **functions and services** including their relationships from an architectural viewpoint
- functions are represented **independent from actors and physical implementations** in applications, systems and components
- functions are derived by extracting **use case functionality** which is independent from actors

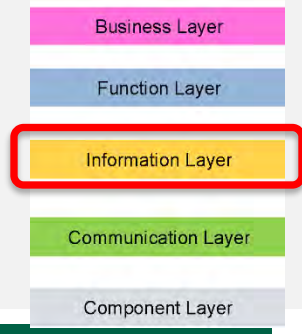
- *e.g. (TSA)*
  aggregation based on combining constraints and flexibility



Information Layer

- describes info used / exchanged between functions, services and components
- contains **information objects** and underlying **canonical data models**
- these represent **common semantics** for functions and services in order to allow an **interoperable information exchange** via communication means

- *e.g. (TSA) device flexibility unlocked via openADR standard*



Communication Layer

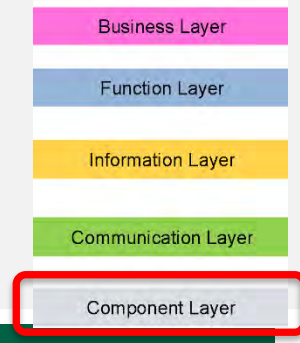
- emphasis is to describe **protocols and mechanisms** for interoperable info exchange between components
  - in the context of the underlying use case, function or service and related information objects or data models

- *e.g. (TSA) HEMS communicates with appliances via ZigBee*

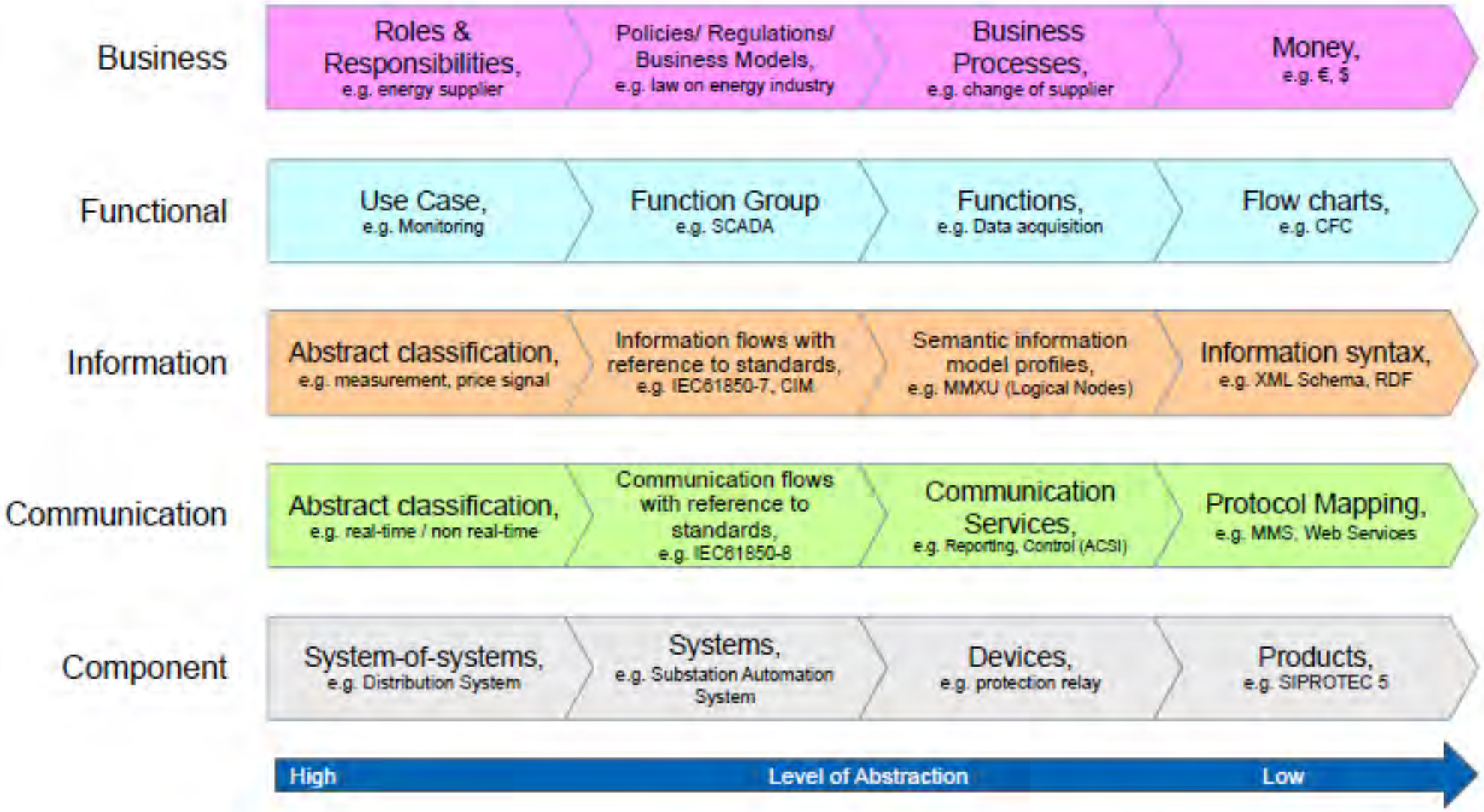
Component Layer

- emphasis on **physical distribution** of components in smart grid context
- includes system actors, applications, power system equipment (typically located at process and field level), protection and telecontrol devices, network infrastructure (wired / wireless communication connections, routers, switches, servers) and any kind of computers

- *e.g. (TSA) optimisation algorithm is executed in the cloud*



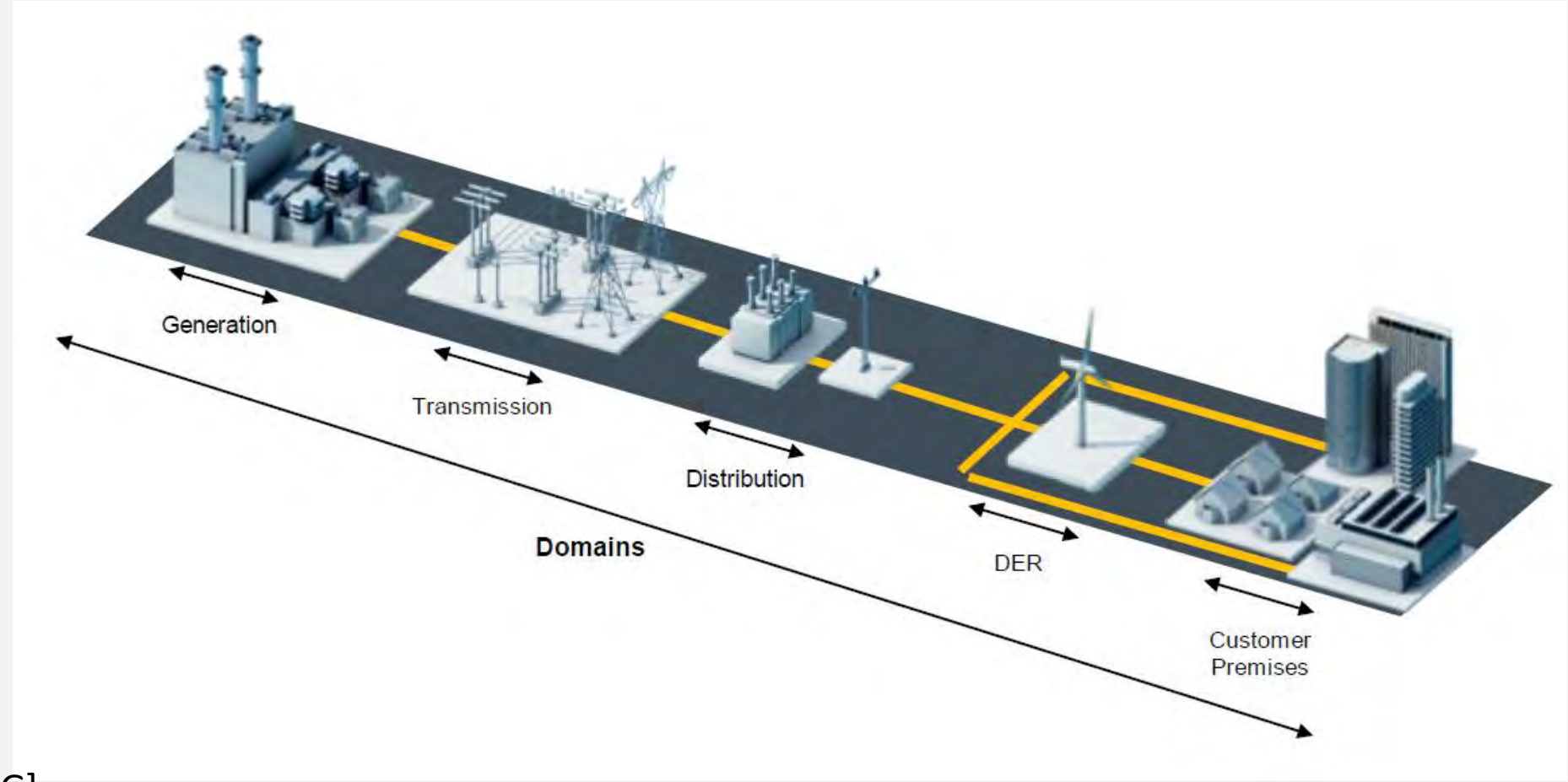
Layers from high to low abstraction level



source: Heiko Englert, Siemens, IEC/TC 57 secretary

x-axis: SGAM domains

- *domains* describe energy conversion chain



source: [SG-CG]

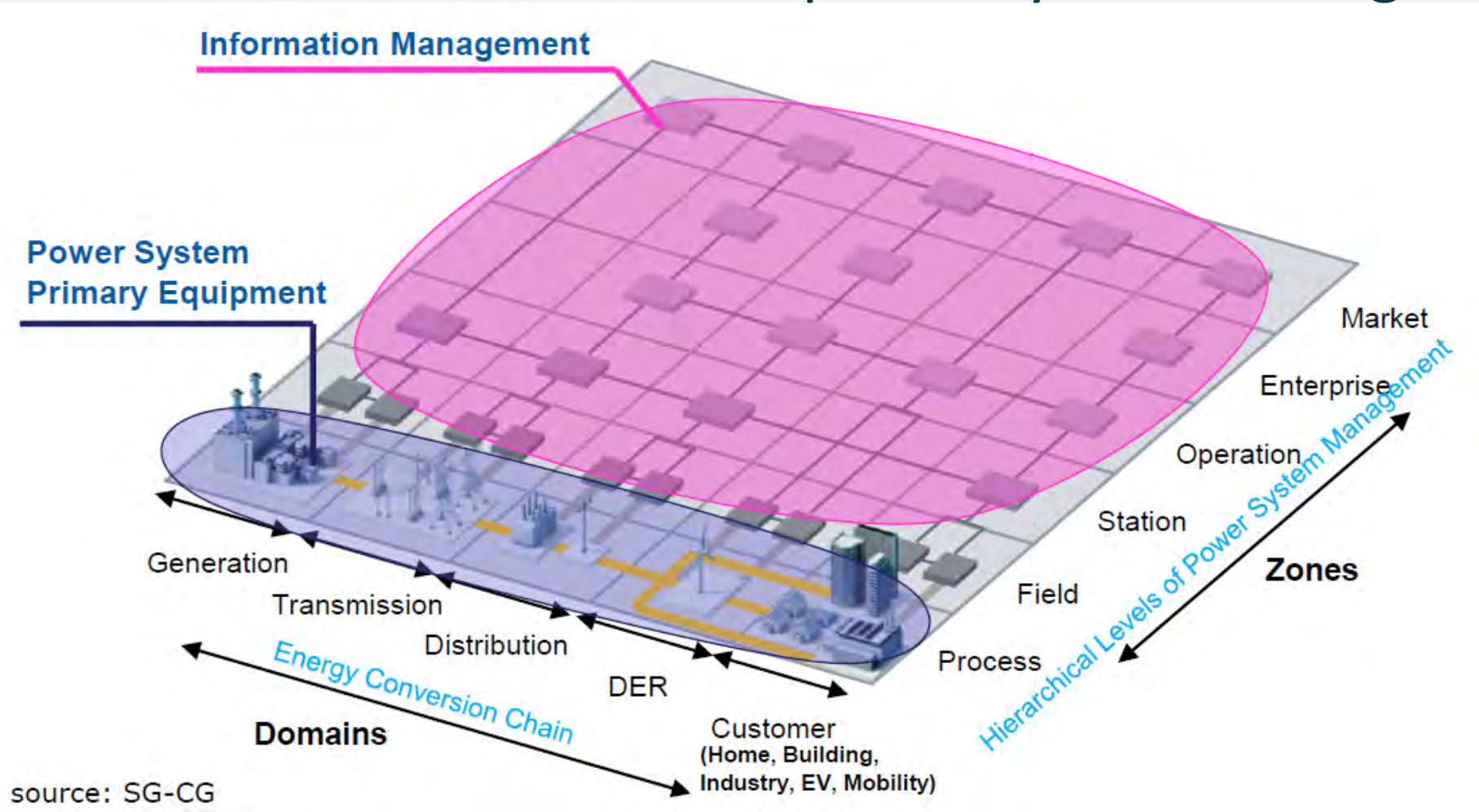
Smart grid plane – x-axis: domains

Table 2: SGAM Domains

Domain	Description
Bulk Generation	Representing generation of electrical energy in bulk quantities, such as by fossil, nuclear and hydro power plants, off-shore wind farms, large scale solar power plant (i.e. PV, CSP)– typically connected to the transmission system
Transmission	Representing the infrastructure and organization which transports electricity over long distances
Distribution	Representing the infrastructure and organization which distributes electricity to customers
DER	Representing distributed electrical resources directly connected to the public distribution grid, applying small-scale power generation technologies (typically in the range of 3 kW to 10.000 kW). These distributed electrical resources may be directly controlled by DSO
Customer Premises	Hosting both - end users of electricity, also producers of electricity. The premises include industrial, commercial and home facilities (e.g. chemical plants, airports, harbors, shopping centers, homes). Also generation in form of e.g. photovoltaic generation, electric vehicles storage, batteries, micro turbines… are hosted

y-axis: zones

- *zones* describe hierarchical levels of power system management



Smart grid plane – y-axis: zones

Table 3: SGAM Zones

Zone	Description
Process	Including the physical, chemical or spatial transformations of energy (electricity, solar, heat, water, wind …) and the physical equipment directly involved. (e.g. generators, transformers, circuit breakers, overhead lines, cables, electrical loads any kind of sensors and actuators which are part or directly connected to the process,…).
Field	Including equipment to protect, control and monitor the process of the power system, e.g. protection relays, bay controller, any kind of intelligent electronic devices which acquire and use process data from the power system.
Station	Representing the areal aggregation level for field level, e.g. for data concentration, functional aggregation, substation automation, local SCADA systems, plant supervision…
Operation	Hosting power system control operation in the respective domain, e.g. distribution management systems (DMS), energy management systems (EMS) in generation and transmission systems, microgrid management systems, virtual power plant management systems (aggregating several DER), electric vehicle (EV) fleet charging management systems.
Enterprise	Includes commercial and organizational processes, services and infrastructures for enterprises (utilities, service providers, energy traders …), e.g. asset management, logistics, work force management, staff training, customer relation management, billing and procurement…
Market	Reflecting the market operations possible along the energy conversion chain, e.g. energy trading, mass market, retail market..

Smart grid architecture model (SGAM) framework – 3D

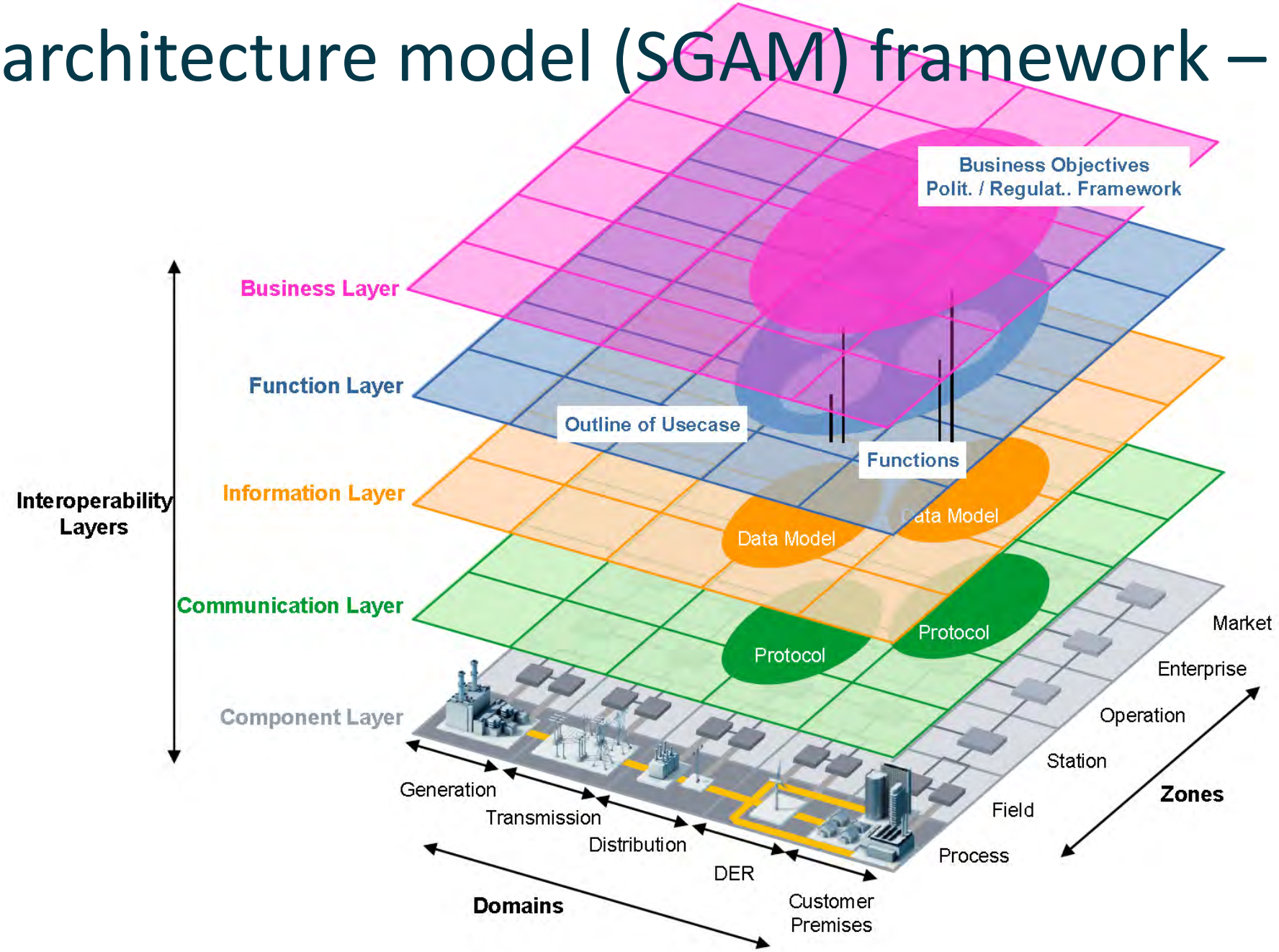


Figure 8: SGAM framework

SGAM usage

- smart grid applications imply
  - many actors
  - many devices and components from multiple vendors
  - many protocols

- SGAM use: draw use case at different abstraction levels
  - at different interoperability levels
  - over different domains and zones

SGAM example: use case distribution grid monitoring (WGSP-0600)

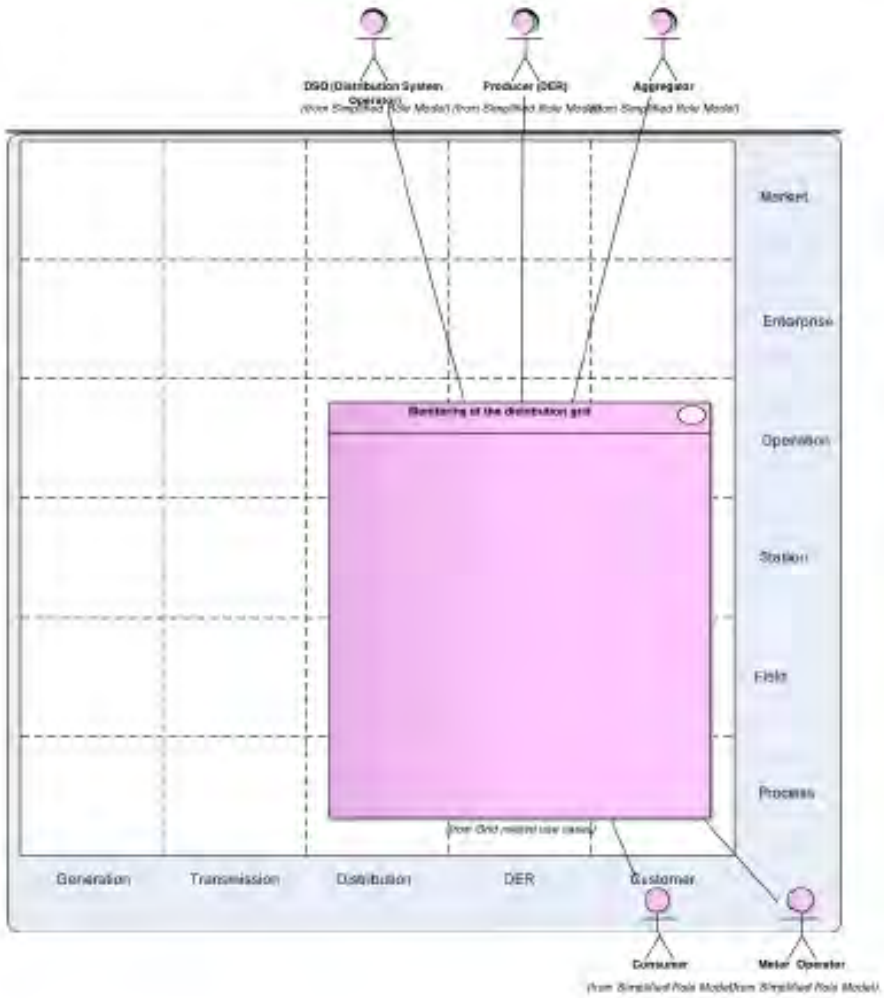
Business Layer
- Different roles involved: DSO, aggregator, customer, ..

Function Layer
- Several ways how monitoring information can be collected (sec. substations, AMI, CEMS, ..)

Information Layer
- power quality data, power flow, protection
- surveillance of substation
- Data from DERs, customers and meters

Communication Layer
- Connectivity

Component Layer
- Sensors & actors at lower voltage levels

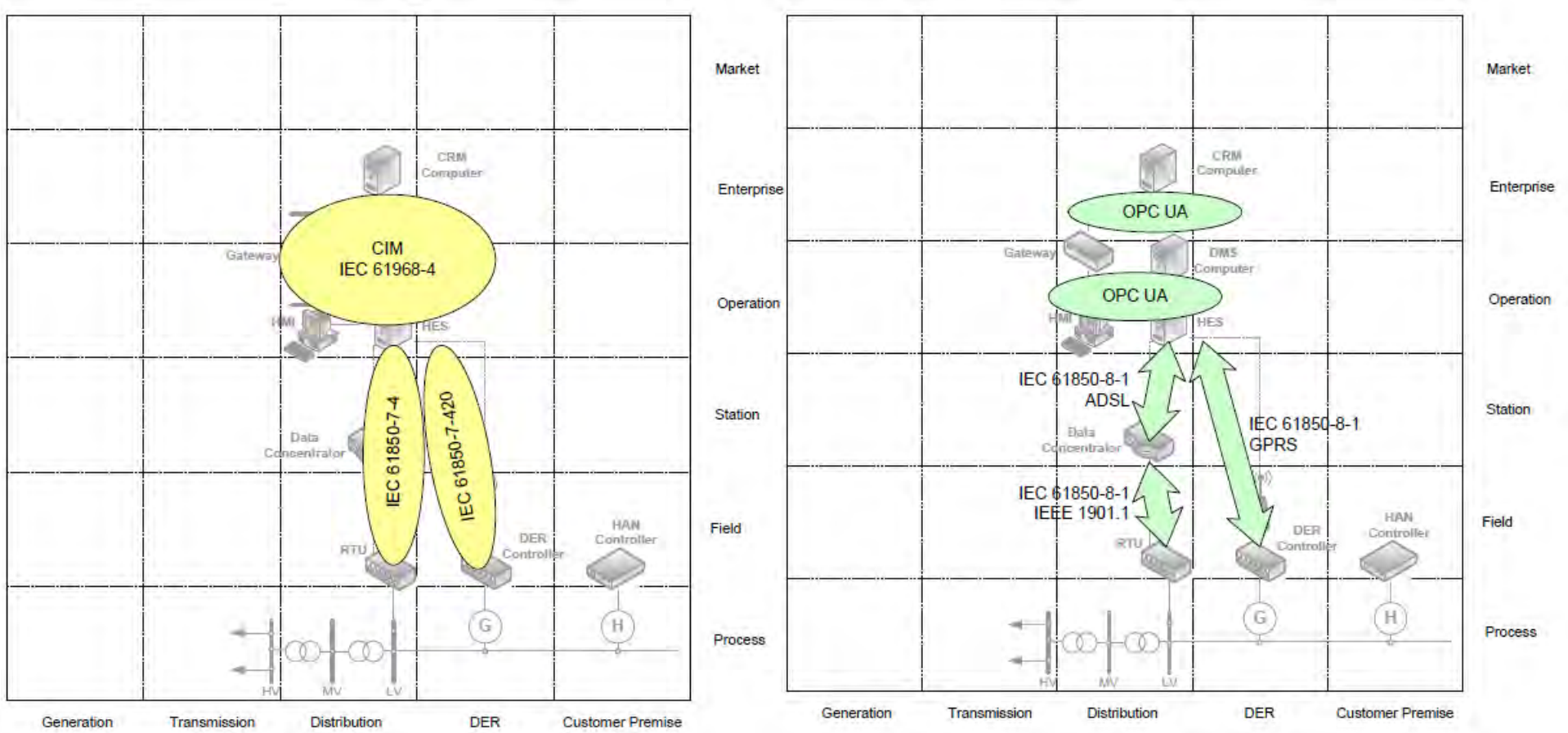


source: SG-CG

SGAM use case (ctd) distribution system monitoring

information layer

communication layer



source: SG-CG

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Example – V/VAr control use case in *function layer*

- *goal:* control reactive power of DER unit to keep voltage within limits

source: [SG-CG]

Figure 17: Flexibility for assignment of function "Volt/Var Control" to SGAM segments, case A - in operation zone, case B - in field zone

Example SGAM: ‘control reactive power of DER unit’

- name, scope and objective

Table 6: Scope and Objective

Scope and Objectives of Use Case	
Related business case	Operation of distribution grid
Scope	Monitor voltage level in distribution grid, control reactive power of DER unit, volt/var control of distribution grid,
Objective	Monitor and control voltage level of distribution grid in tolerated limits

Example ctd: use case diagram

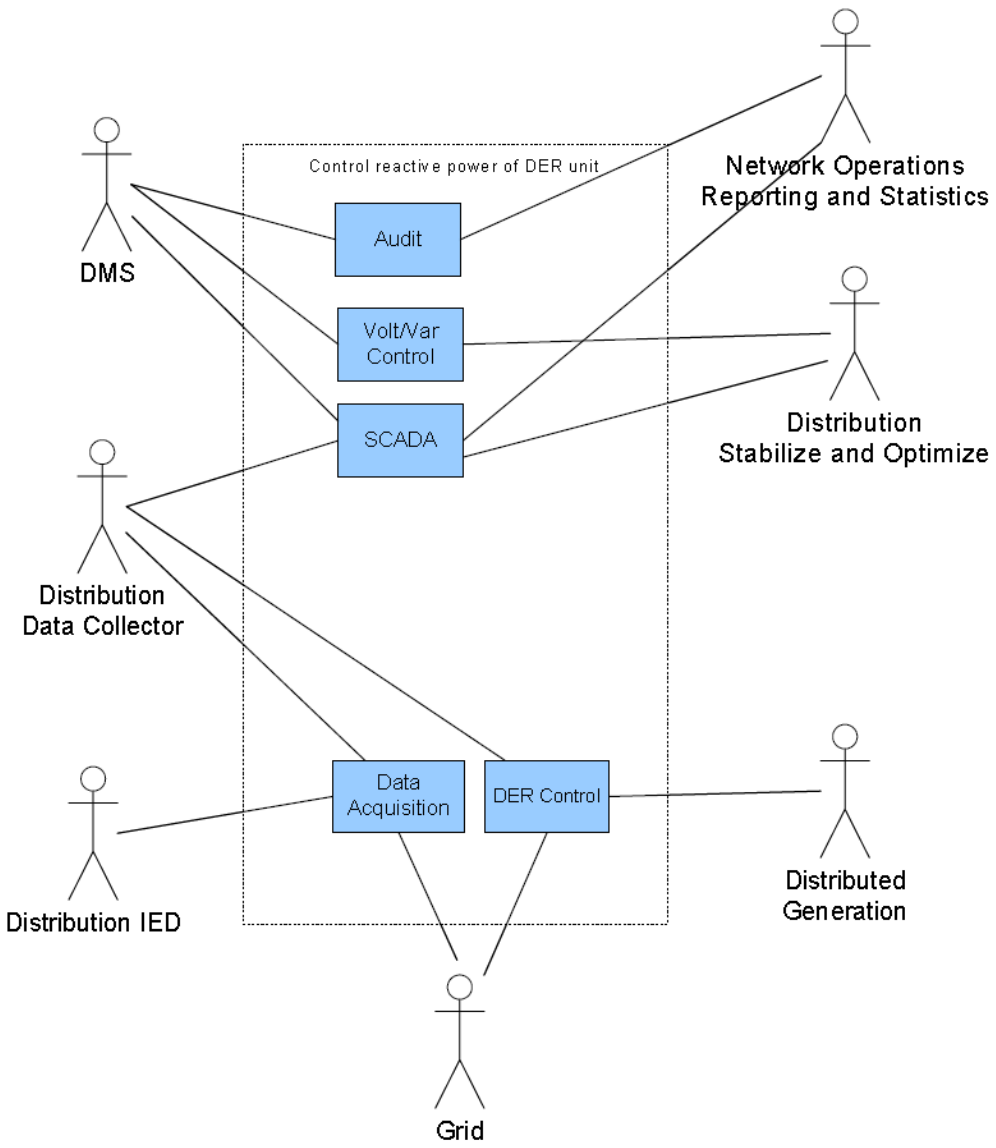

Figure 24: Use Case Diagram for "*Control reactive power of DER unit*"

Example ctd: actors

Actors			
Grouping (Community)		**Group Description**	
Actor Name see Actor List	**Actor Type** see Actor List	**Actor Description** see Actor List	**Further information specific to this Use Case**
Grid	System	Power Distribution system	
Distribution-IED	Device	Intelligent Electric Device (IED) is a communications-enabled controller to monitor and control automated devices in distribution which communicates with Distribution SCADA or other monitoring/control applications, as well as distributed capabilities for automatic operations in a localized area based on local information and on data exchange between members of the group. Operations such as such as tripping circuit breakers if they sense voltage, current, or frequency anomalies.	
Distributed Generation	Device	Distributed Generation, also called Distributed Energy Resources (DER), includes small-scale generation or storage of whatever form. This is in contrast to centralized or bulk generation and/or storage of electricity. These generation facilities are part of Demand/Reponse programs and may be dispatchable resources. The primary distictions between Distributed Generation (DG) and Bulk Generation is: Bulk Generation is attached via Transmission facilities, output is sold in wholesale markets, provides base load; DG is sited on a Customer permises attached to the Distribution grid, output is Retail (unless sold via an aggregator), can provide ancillary services. These generation facilities include but are not limited to Photvoltaic panel (PV), micro-hydro, mindmill, Plug-in Hybrid/Electric Cars (PHEV), and potentially Fuel cells. These facilites are usually not scheduled but can be dispatched.	

Actors (ctd.)

Distribution Data Collector	Device	A data concentrator bringing data from multiple sources and putting it into different form factors.	
Distribution Stabilize and Optimize	Application	Performed by actors to ensure the network is operating within appropriate tolerances across the system. They gather information for control decisions that ensure reliable, proper operations (stability) and more efficient operations (optimization). Measurement and control form a feedback loop that allows grid operators to stabilize the flow of energy across the electric network or safely increase the load on a transmission path.	
Distribution Management System	Application	A suite of application software that monitors and controls the distribution system equipment based on computer-aided applications, market information, and operator control decisions.	
Network Operations Reporting and Statistics	Application	Operational Statistics and Reporting actors archive on-line data and to perform feedback analysis about system efficiency and reliability.	

Example ctd: conditions

Table 8: Preconditions, Assumptions, Post condition "*Control reactive power of DER unit*"

Use Case Conditions			
Actor/System/Information/Contract	***Triggering Event***	***Pre-conditions***	***Assumption***
Distribution Management System		• The Grid is continuously monitored • The Grid topology is known and reflects the real topology • The Grid energy path is known and reflects the real path (effective status of remote monitored and controllable switches)	
Distribution-IED		The device is up and running	
Distributed Generation		The DER is connected to the grid and injects active and reactive power	
Distribution Data Collector		The device is up and running	
Distribution Stabilize and Optimize		The application is up and running	
Distribution Management System		The application is up and running	
Network Operations Reporting and Statistics		The application is up and running	

Example ctd: analysis

Table 9: Step by Step Analysis of Use Case "*Control reactive power of DER unit*"

Scenario Conditions					
No.	**Scenario Name**	**Primary Actor**	**Triggering Event**	**Pre-Condition**	**Post-Condition**
4.1	Data Acquisition	Distribution IED	Periodically		
4.2	SCADA	DMS	Periodically		
4.3	Voltage/Var Control	Distribution Stabilize and Optimize	Voltage Measurement exceeded threshold		
4.4	DER Control	DMS	Control value, equipment id, received		
4.5	Audit	DMS	Control action		

Example ctd: use case steps

Table 10: Use Case Steps "*Control reactive power of DER unit*"

Step #	Triggering Event	Actor *What actor, either primary or secondary is responsible for the activity in this step?*	Description of the activity *Describe the actions that take place in this step including the information to be exchanged. The step should be described in active, present tense*	Information producer	Information Receiver	Information exchanged	Additional Notes *Elaborate on any additional description or value of the step to help support the descriptions. Short notes on architecture challenges, etc. may also be noted in this column*
Data Acquisition"							
1a	Periodically	Distribution IED	Distribution IED acquires analogue voltage measurement	Grid	Distribution IED	Analogue Voltage Measurement	
2	Periodically	Distribution IED	Distribution IED transmits voltage measurement	Distribution IED	Distribution Data Collector	Voltage Measurement	
3	Periodically	Distribution Data Collector	Distribution Data Collector transmits voltage measurement to DMS system.	Distribution Data Collector	DMS	Voltage Measurement	
Scada							
4	Periodically	DMS	The DMS System collects data from the grid, reformates the data and complements it with additional relevant information , distributes the data to DMS applications	DMS	Network Operations Reporting & Statistics, Distribution Stabilize and Optimize	Voltage Measurement, location, topology information	
Voltage/Var Control							
5	Voltage Measurement exceeded threshold	Distribution Stabilize and Optimize	Distribution Stabilize and Optimize application detects a threshold violation of voltage	Distribution Stabilize and Optimize	Distribution Stabilize and Optimize	Violation information	
6	Threshold Violation	Distribution Stabilize and Optimize	Distribution Stabilize and Optimize application starts Voltage/Var calculation	Distribution Stabilize and Optimize	Distribution Stabilize and Optimize	Start of voltage/Var calculation	
7	Start voltage Var calculation	Distribution Stabilize and Optimize	Distribution Stabilize and Optimize application calculates control value and identifies equipment to be controlled and transmits value to DMS	Distribution Stabilize and Optimize	DMS	Control value, equipment id	

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Use case steps (ctd)

Step #	Triggering Event	Actor *What actor, either primary or secondary is responsible for the activity in this step?*	Description of the activity *Describe the actions that take place in this step including the information to be exchanged. The step should be described in active, present tense*	Information producer	Information Receiver	Information exchanged	Additional Notes *Elaborate on any additional description or value of the step to help support the descriptions. Short notes on architecture challenges, etc. may also be noted in this column*
DER Control							
8	Control value, equipment id, received	DMS	DMS reformats control value and equipment id and transmits controllable setpoint to Distribution Data Collector	DMS	Distribution Data Collector	Controllable setpoint	
9	Controllable setpoint received	Distribution Data Collector	Distribution Data Collector device forwards information to Distributed Generation device	Distribution Data Collector	Distributed Generation	Controllable setpoint	
10	Controllable setpoint received	Distributed Generation	Distributed Generation device updates its operation parameters according to setpoint	Distributed Generation	Distributed Generation	Operation parameter	
11	Operation parameter update	Distributed Generation	Distributed Generation device verifies updated operation mode and acknowledges parameter change	Distributed Generation	Distribution Data Collector	Acknowledge information	
12	Acknowledge information received	Distribution Data Collector	Distribution Data Collector device forwards information to DMS	Distribution Data Collector	DMS	Acknowledge information	
Audit							
13	Control action	DMS	DMS application posts control action to Network Operations Reporting & Statistics application	DMS	Network Operations Reporting & Statistics	Control action	
14	Control action	Network Operations Reporting & Statistics	Network Operations Reporting & Statistics application documents control action	Network Operations Reporting & Statistics	Network Operations Reporting & Statistics	Control action	

Example ctd: mapping to domain and zones

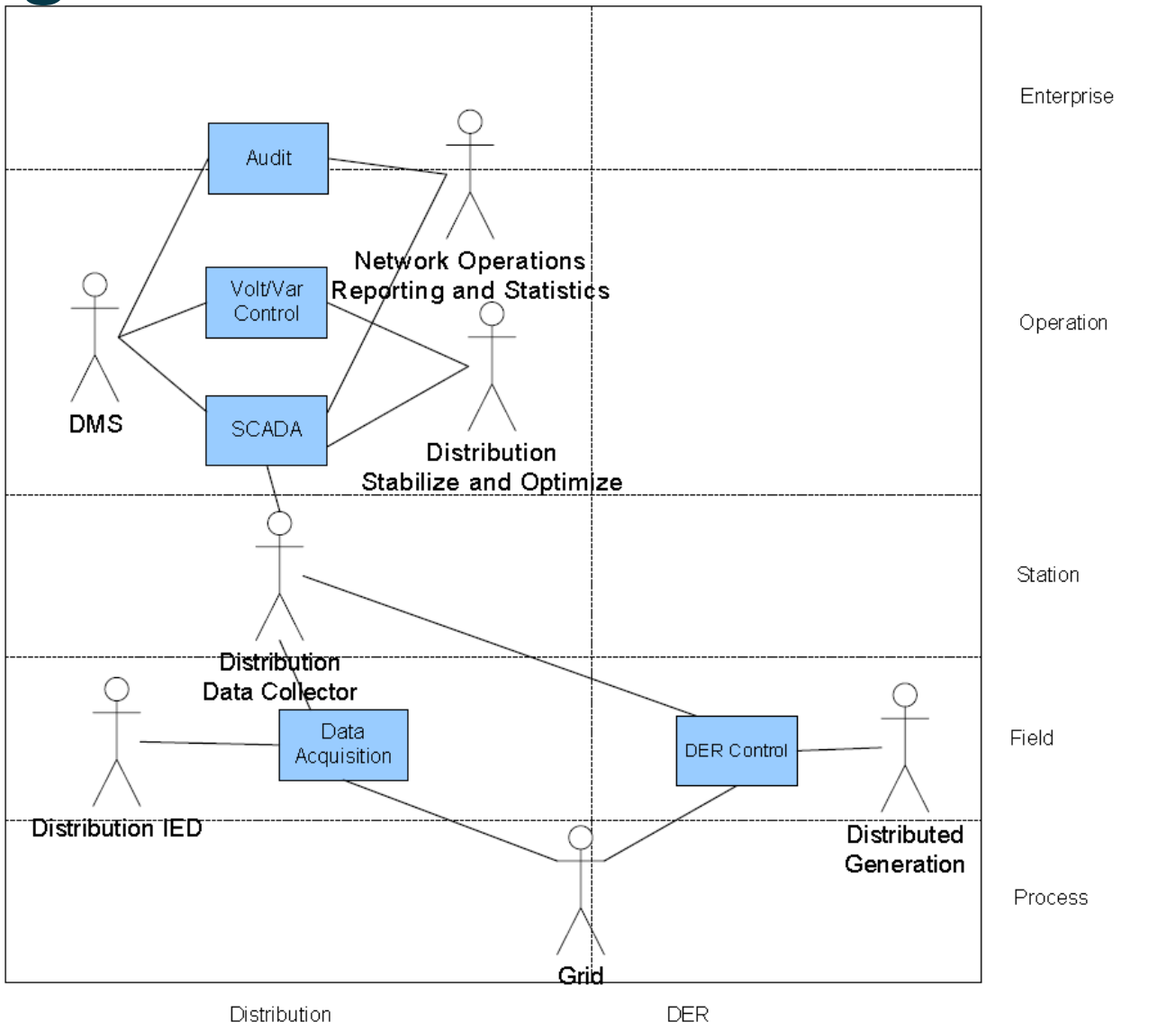


Figure 25: Actors and sub use cases mapped to domains and zones, "*Control reactive power of DER unit*"

Example ctd: component layer

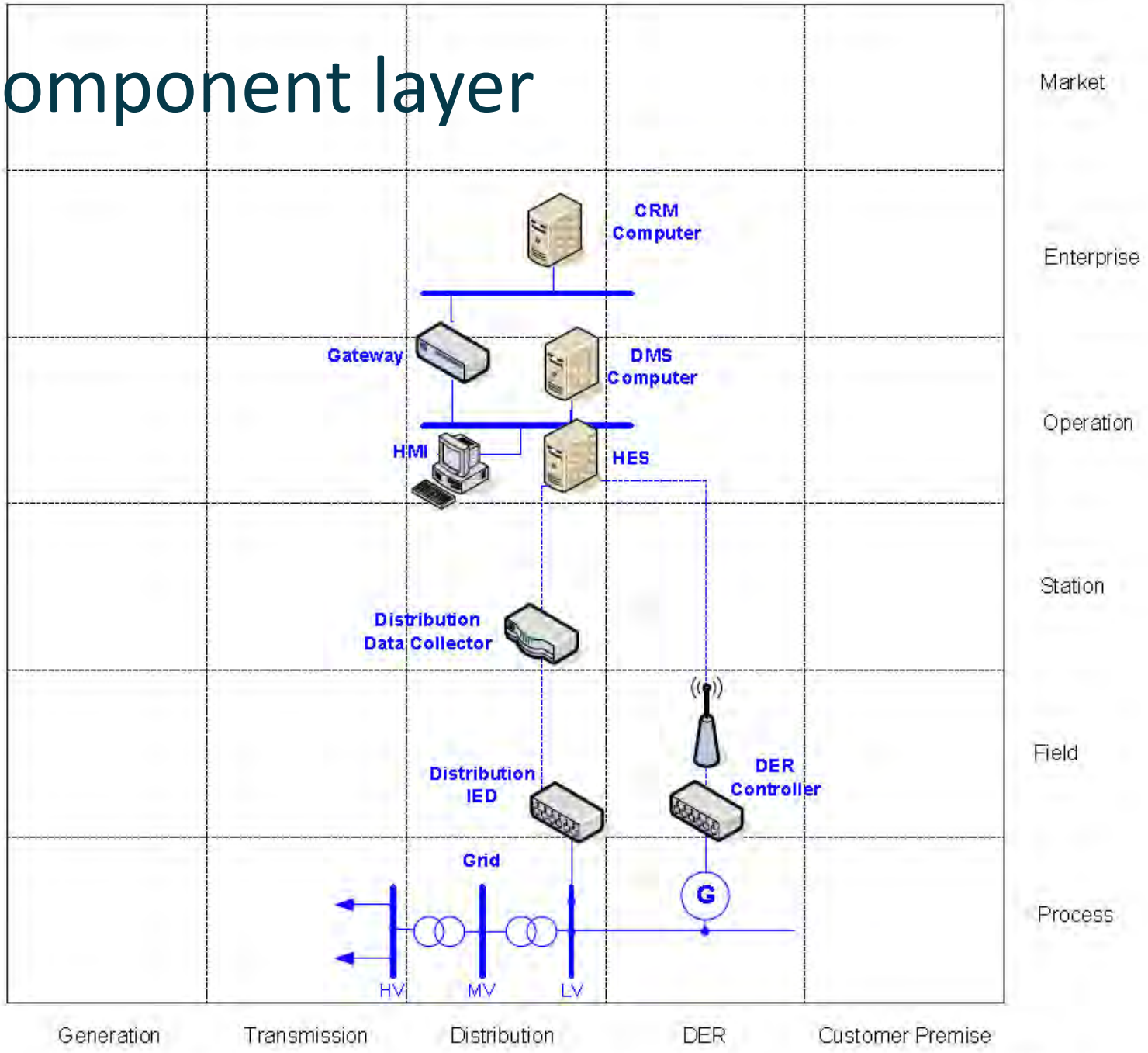


Figure 26: Component Layer "*Control reactive power of DER unit*"

Example ctd: function layer

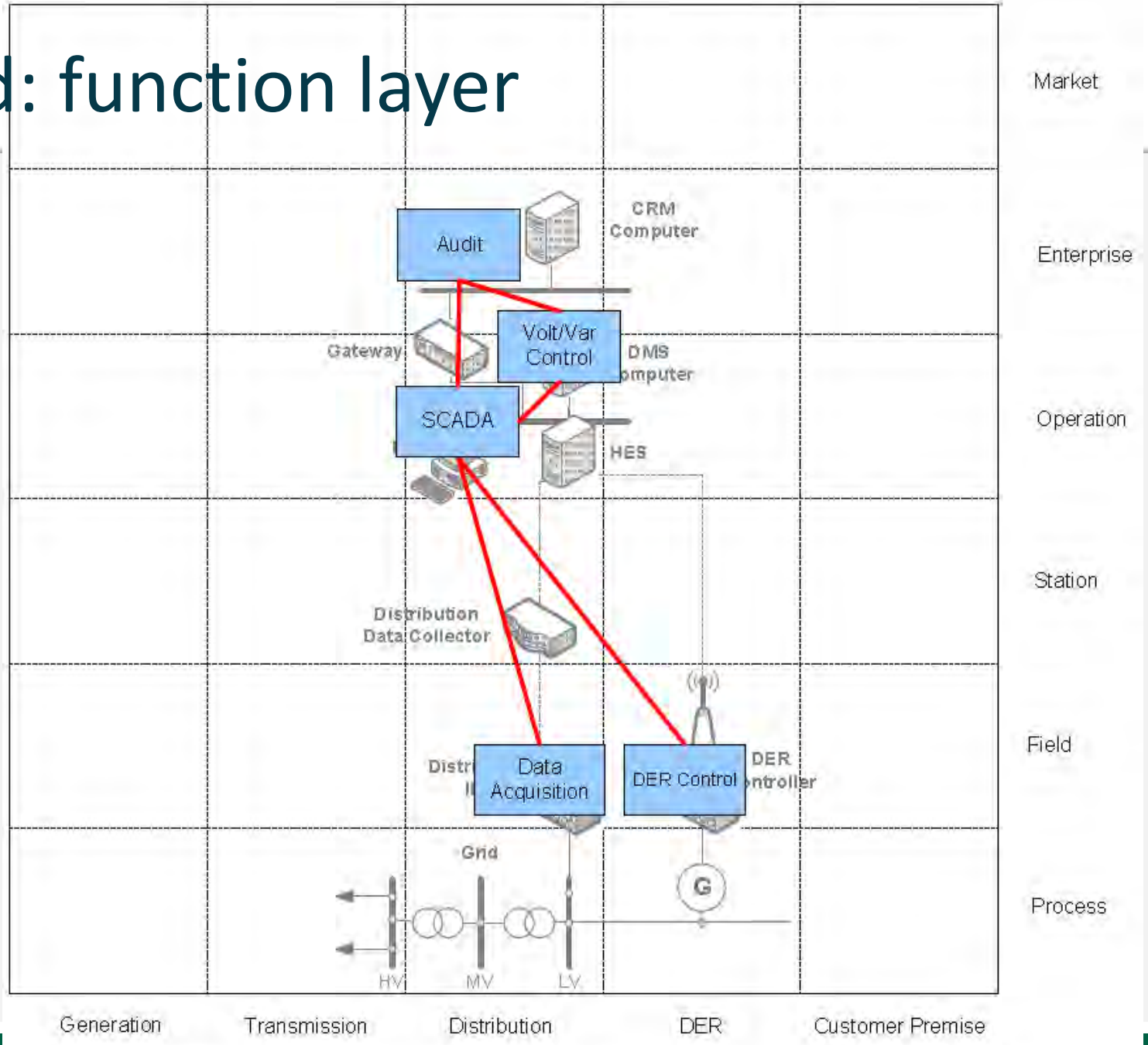


Figure 28: Function Layer "*Control reactive power of DER unit*"

Example ctd: information layer / business context

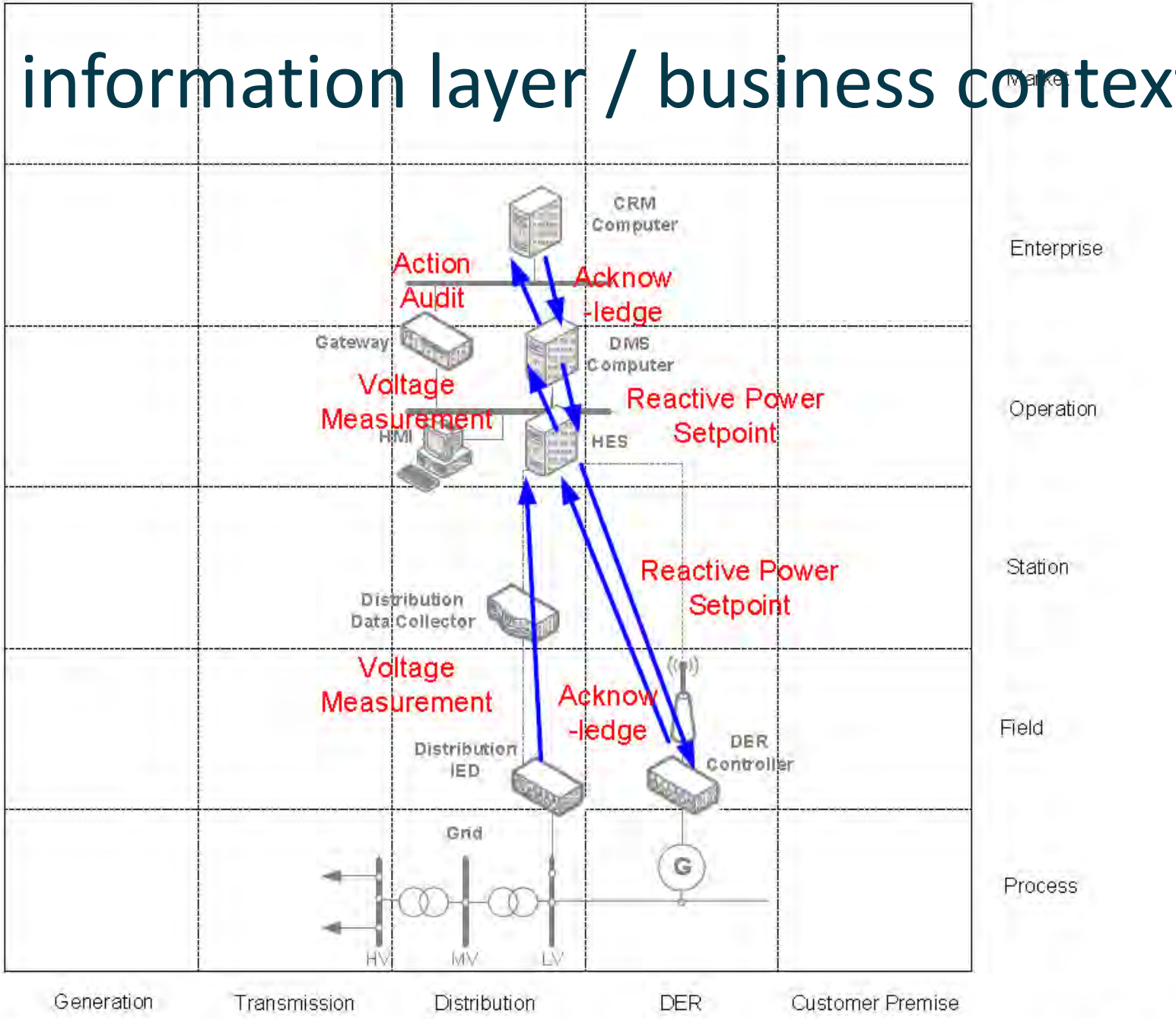


Figure 29: Information Layer / Business Context view, "*Control reactive power of DER unit*"

Example ctd: information model / data model

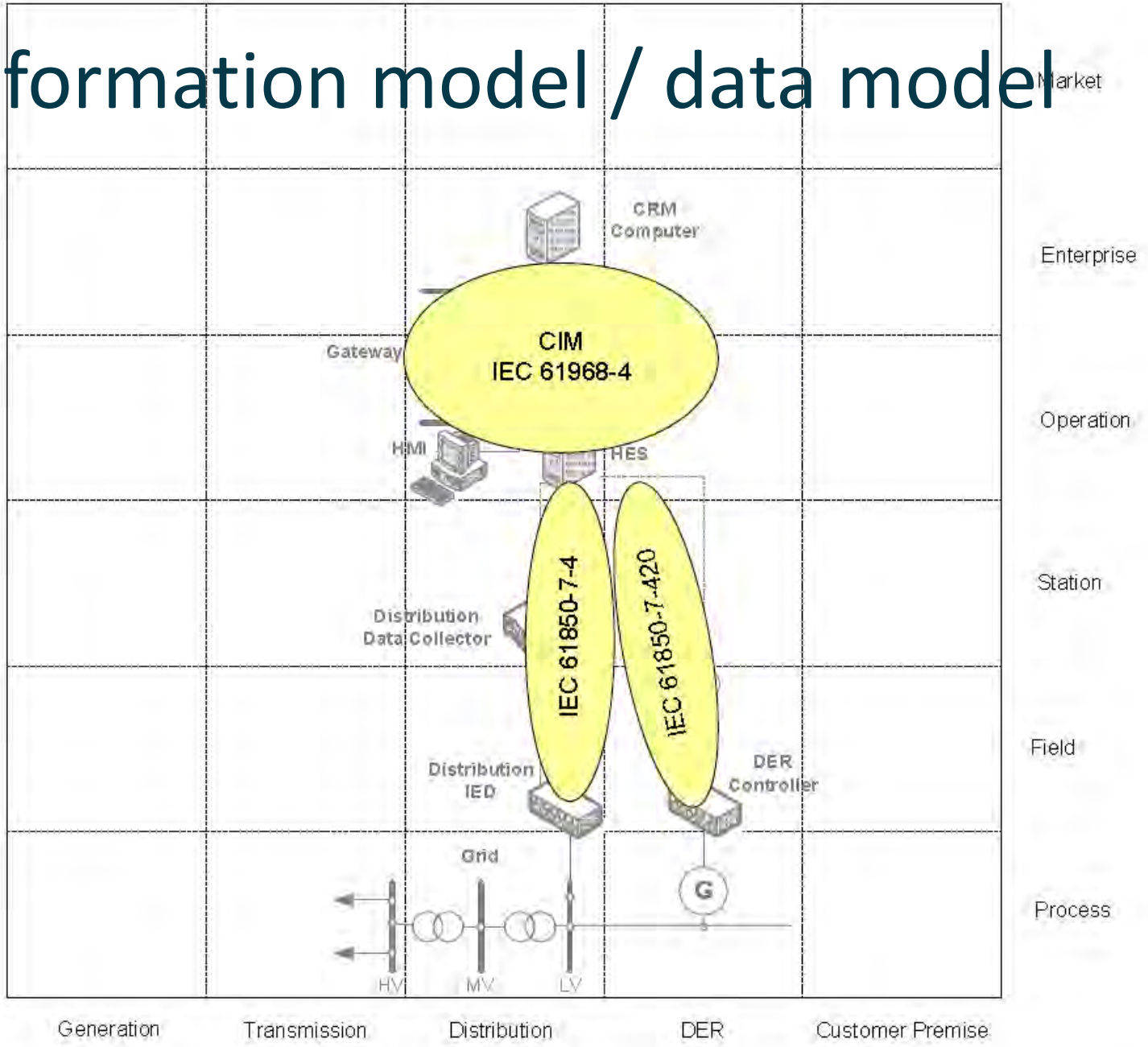


Figure 30: Information Layer / Canonical Data Model view, "*Control reactive power of DER unit*"

Example ctd: communication layer

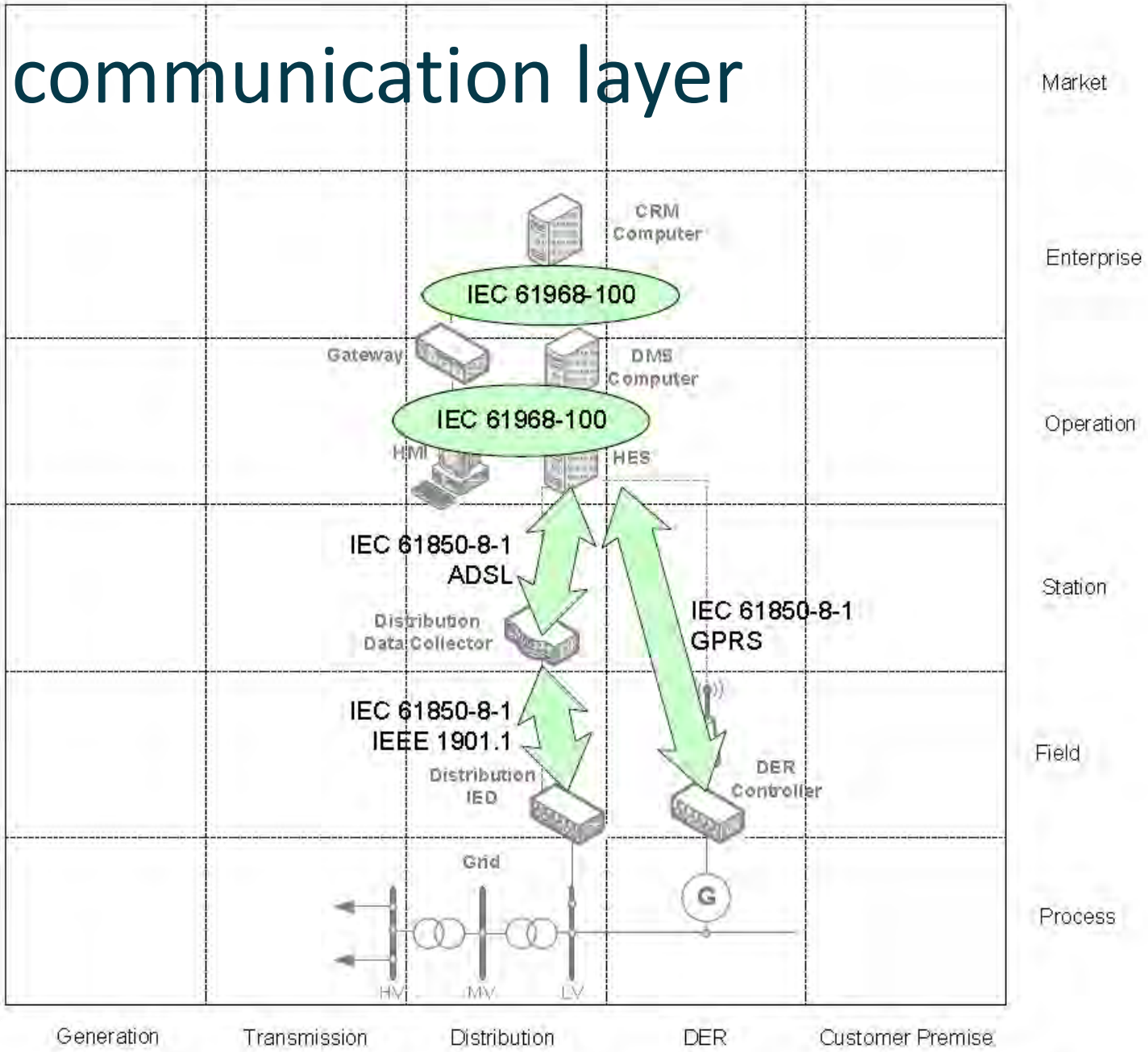


Figure 31: Communication Layer "*Control reactive power of DER unit*"

Outline

- Three step approach for flexibility management
- SGAM - smart grid architecture model
  - structure
  - example
  - layers' contents

SGAM function layer

- maps smart grid functions
  - to zones and domains
- abbreviations
  - AMI: automated meter infrastructure
  - EMS: energy management system
  - VPP: virtual power plant
  - DER: distributed energy resource
  - FACTS: flexible ac transmission system
  - SCADA: supervisory control and data acquisition system
  - WAMS: wide area measurement system

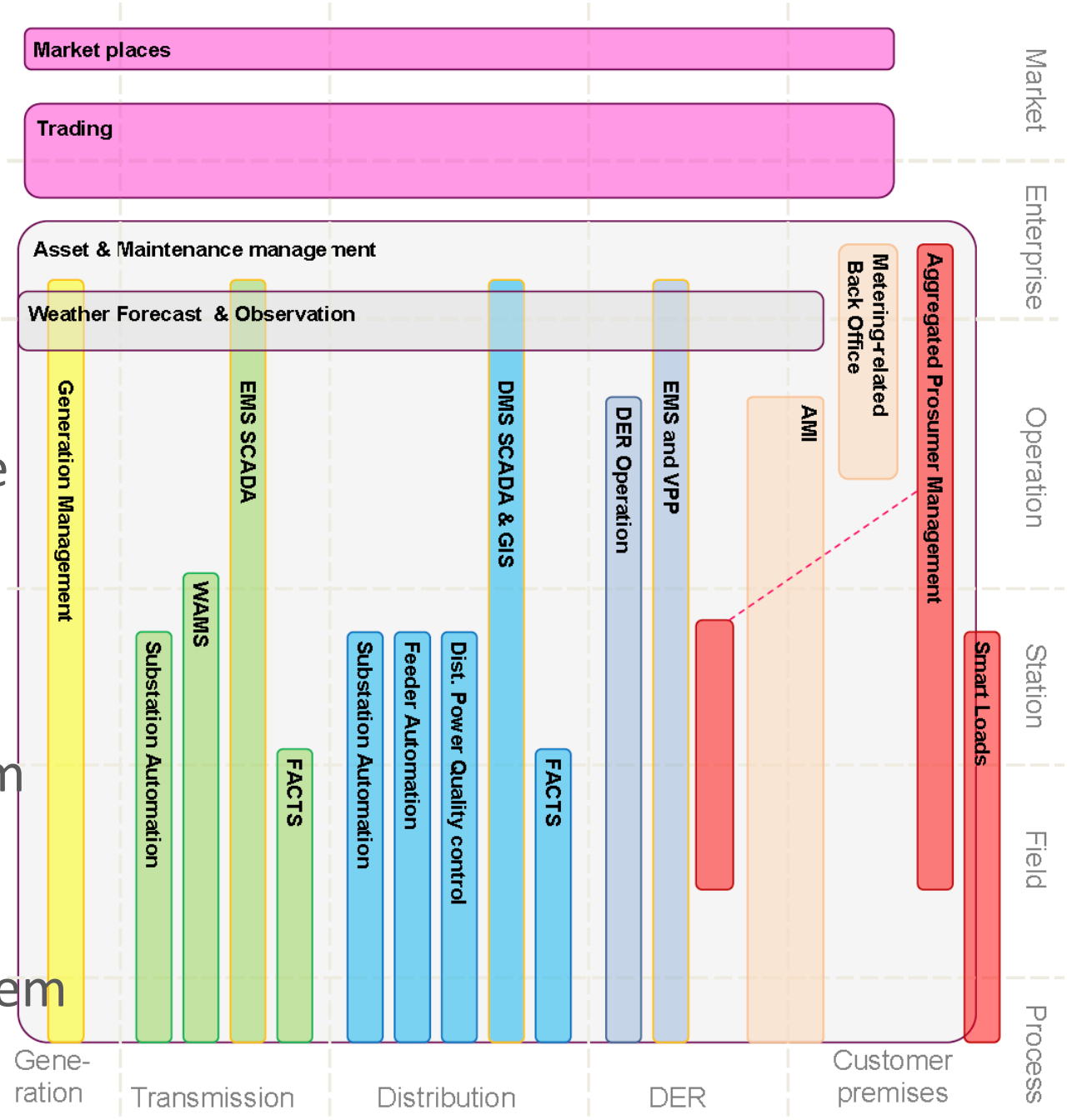
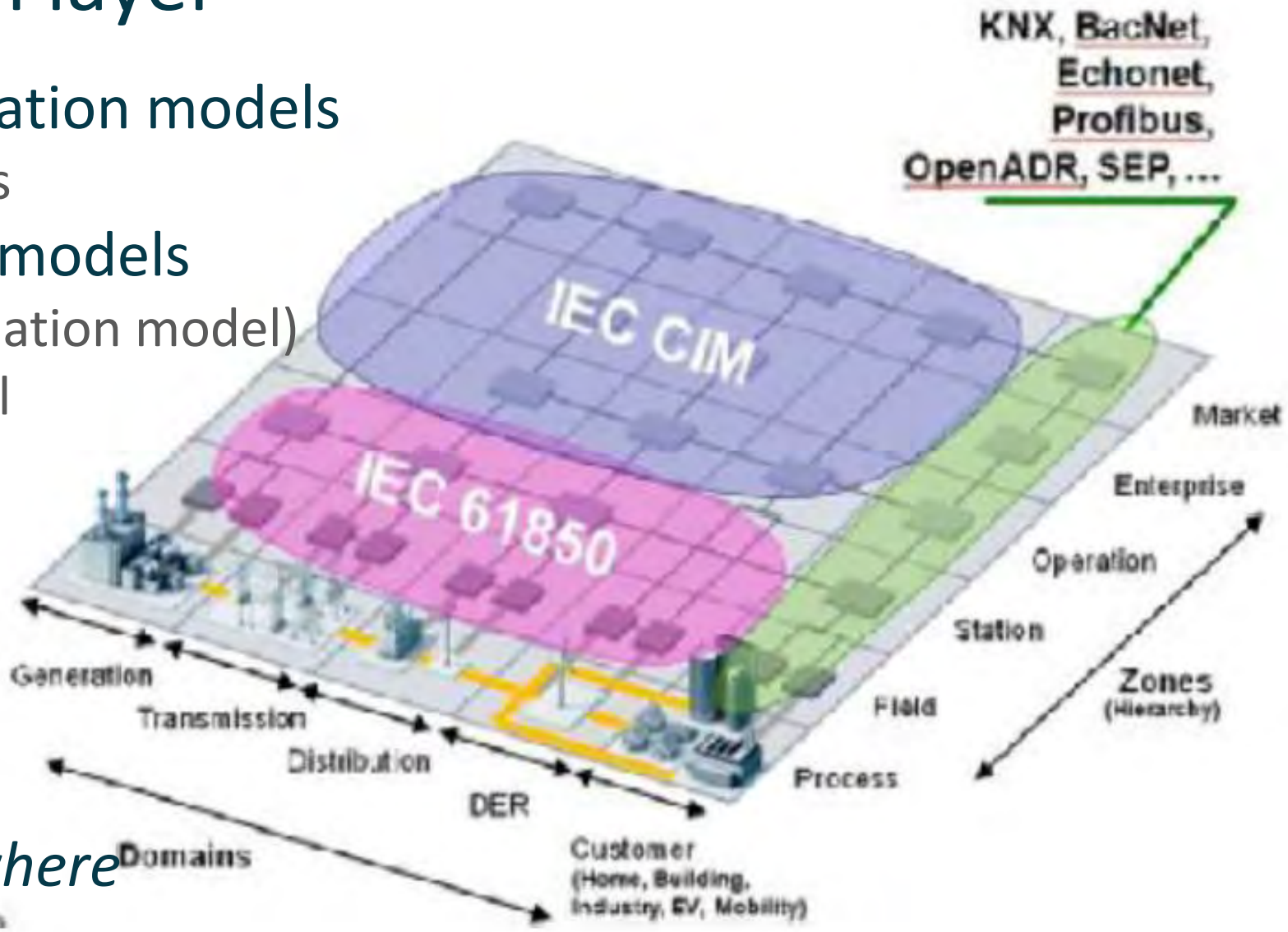


Figure 18: Overview on Smart Grid function groups derived from Smart Grid Systems

SGAM information layer

- maps data and information models
  - to zones and domains
- two mostly used data models
  - CIM (common information model)
  - IEC 61850 data model



- *should overlap somewhere*

SGAM communication layer

- maps communication networks
  - to zones and domains

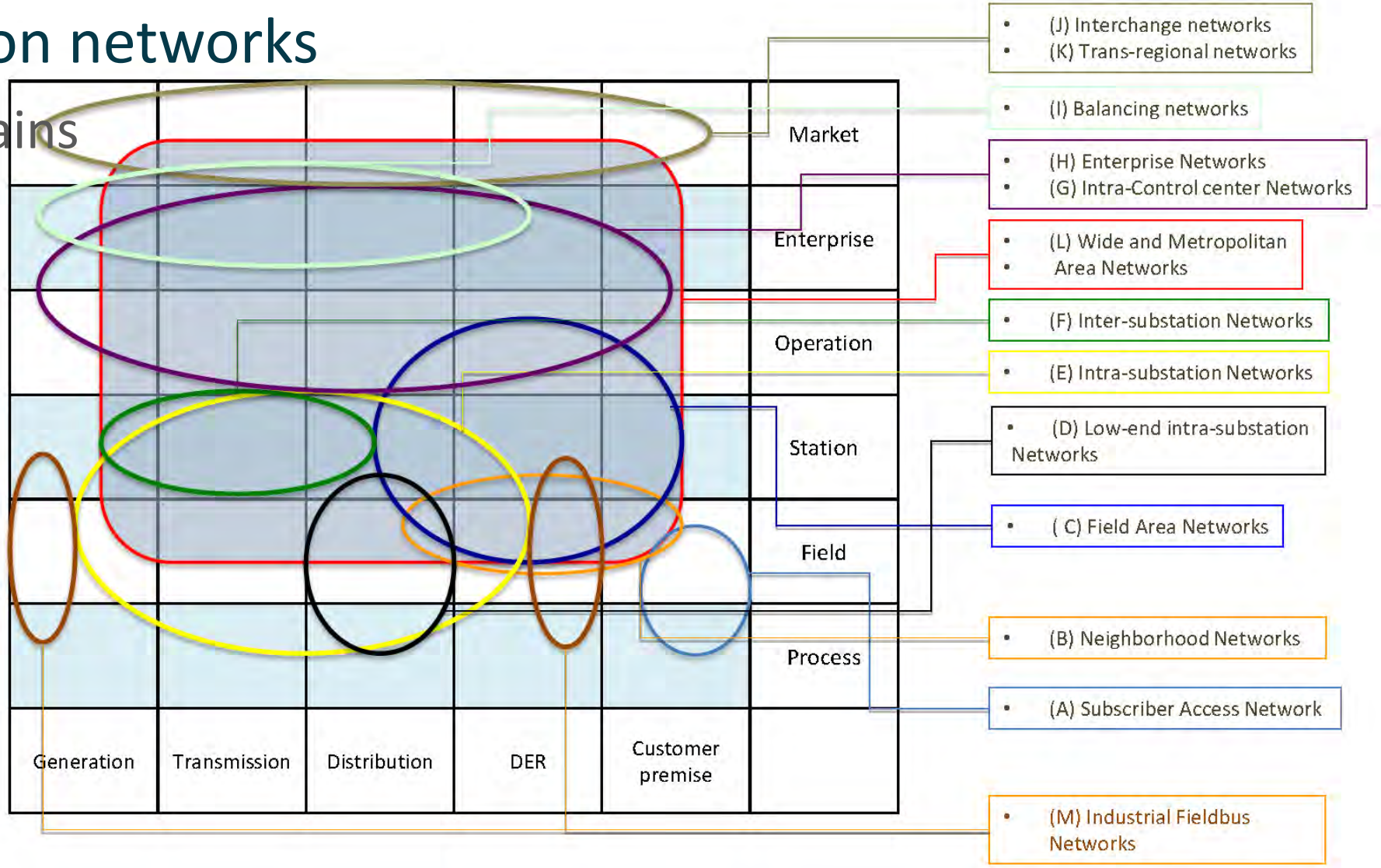


Figure 20: Mapping of communication networks on SGAM Communication Layer

Communication networks & comm. technologies

Table 5: Applicability statement of the communication technologies to the smart grid sub-networks

	A Subscriber access network	B Neighbourhood Network	C Field area	D Low-end intra substation	E Intra-substation	F Inter-substation	G Intra control centre	G intra data centre	H Enterprise	I Balancing	J Interchange	K Trans regional	K Trans national	L WAN	M Industrial Fieldbus
Narrow band PLC (Medium and Low voltage)	x	x	x												
Narrow band PLC (High and very High voltage)					x	x									
Broadband PLC	x	x													
IEEE 802.15.4	x	x	x												
IEEE 802.11	x	x		x	x										
IEEE 802.3/1				x	x		x	x	x						x
IEEE 802.16	x	x	x												
ETSI TS 102 887		x	x												
IPv4	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
IPv6	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
RPL / 6LowPan	x	x	x												
IEC 61850		x	x	x	x	x								x	
IEC 60870-5				x	x	x								x	
GSM / GPRS / EDGE	x	x												x	
3G / WCDMA / UMTS / HSPA	x	x					x	x	x	x	x	x	x	x	
LTE/LTE-A	x	x	x	x		x	x	x	x	x	x	x	x	x	
SDH/OTN	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
IP MPLS / MPLS TP	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
EN 13757		x													
DSL/PON	x	x				x								x	

SGAM standardised in IEC technical report

- IEC TR 62357-1:2016
  - Power systems management and associated information exchange – Part 1: Reference architecture
- IEC TR 62357-2:2019
  - Power systems management and associated information exchange – Part 2: Use Cases and role model

IEC

IEC TR 62357-1

Edition 2.0 2016-11

TECHNICAL REPORT

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Power systems management and associated information exchange – Part 1: Reference architecture

2021: Extended SGAM smart energy grid reference architecture model

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IEC SRD 63200:2021

Definition of extended SGAM smart energy grid reference architecture model

SyC Smart Energy | Additional information

Abstract

PREVIEW

IEC SRD 63200:2021(E), which is a Systems Reference Deliverable, defines the framework elements, associated ontology, and modelling methodology for designing the Smart energy Grid Reference Architecture using the Smart Grid Architecture Model (SGAM), with potential expansion to describe the interaction between the grid and heat/gas systems, and including easily understandable examples. This document also provides a machine level representation of the concepts associated with the SGAM in the form of an ontology provided in the form of diagrams in Annex A, as well as in the form of a code component. Thus, this document is associated with a code component presented as a ZIP file package.

1 Scope

This document, which is a Systems Reference Deliverable, defines the framework elements, associated ontology, and modelling methodology for designing the Smart energy Grid Reference Architecture using the Smart Grid Architecture Model (SGAM), with potential expansion to describe the interaction between the grid and heat/gas systems, and including easily understandable examples.

This document also provides a machine level representation of the concepts associated with the SGAM in the form of an ontology provided in the form of diagrams in Annex A, as well as in the form of a code component. Thus, this document is associated with a code component presented as a ZIP file package containing:

- a file describing the content of the package (IECManifest.xml);
- the OWL representation of the ontology of the concepts introduced in this document;
- The same content as the OWL content, but exposed as a series of HTML files, which any browser can open for easy reading.